

## Chapter 6

1.

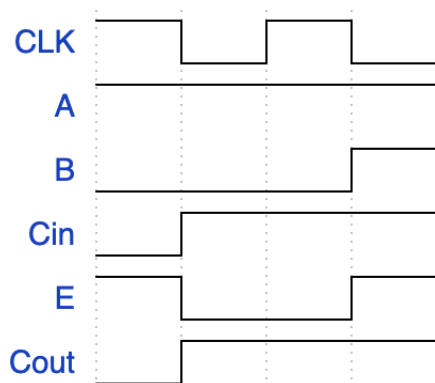
1.a)  $32 \text{ ns}$

$$f = \frac{1}{T}$$
$$= \frac{1}{32 \times 10^{-9}}$$
$$f = 31.25 \text{ MHz}$$

b)  $(7)(8) = 56 \text{ ns}$

$$f = \frac{1}{56 \times 10^{-9}}$$
$$f = 17.86 \text{ MHz}$$

c)



2.

| 2. $A > B$ |           | $A = B$   |           | $A < B$   |           |
|------------|-----------|-----------|-----------|-----------|-----------|
| $A_1 A_0$  | $B_1 B_0$ | $A_1 A_0$ | $B_1 B_0$ | $A_1 A_0$ | $B_1 B_0$ |
| 00         | 00        | 00        | 00        | 00        | 00        |
| 01         | 01        | 01        | 01        | 01        | 01        |
| 11         | 11        | 11        | 11        | 11        | 11        |
| 10         | 10        | 10        | 10        | 10        | 10        |

|    |    |   |   |   |   |
|----|----|---|---|---|---|
| a) | 00 | 0 | 0 | 0 | 0 |
|    | 01 | 1 | 0 | 0 | 0 |
|    | 11 | 1 | 1 | 0 | 1 |
|    | 10 | 1 | 1 | 0 | 0 |

$$= (A_1 \bar{B}_1) + (A_1 \bar{B}_0) + (A_0 \bar{B}_1 \bar{B}_0)$$
  

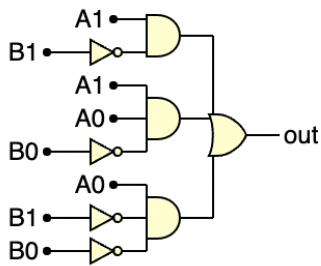
|    |    |   |   |   |   |
|----|----|---|---|---|---|
| b) | 00 | 1 | 0 | 0 | 0 |
|    | 01 | 0 | 1 | 0 | 0 |
|    | 11 | 0 | 0 | 1 | 0 |
|    | 10 | 0 | 0 | 0 | 1 |

$$= (A_1 + \bar{B}_1)(A_0 + \bar{B}_0)(B_1 + \bar{A}_1)(B_0 + \bar{A}_0)$$
  

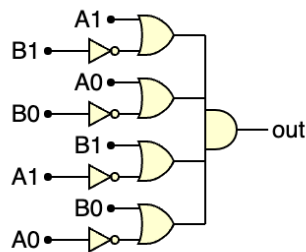
|    |    |   |   |   |   |
|----|----|---|---|---|---|
| c) | 00 | 0 | 1 | 1 | 1 |
|    | 01 | 0 | 0 | 1 | 1 |
|    | 11 | 0 | 0 | 0 | 0 |
|    | 10 | 0 | 0 | 1 | 0 |

$$= (B_1 \bar{A}_1) + (B_1 \bar{A}_0) + (B_0 \bar{A}_1 \bar{A}_0)$$

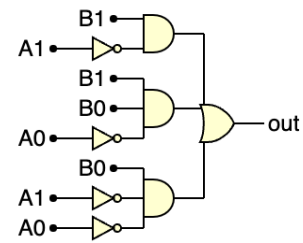
a)  $A > B$



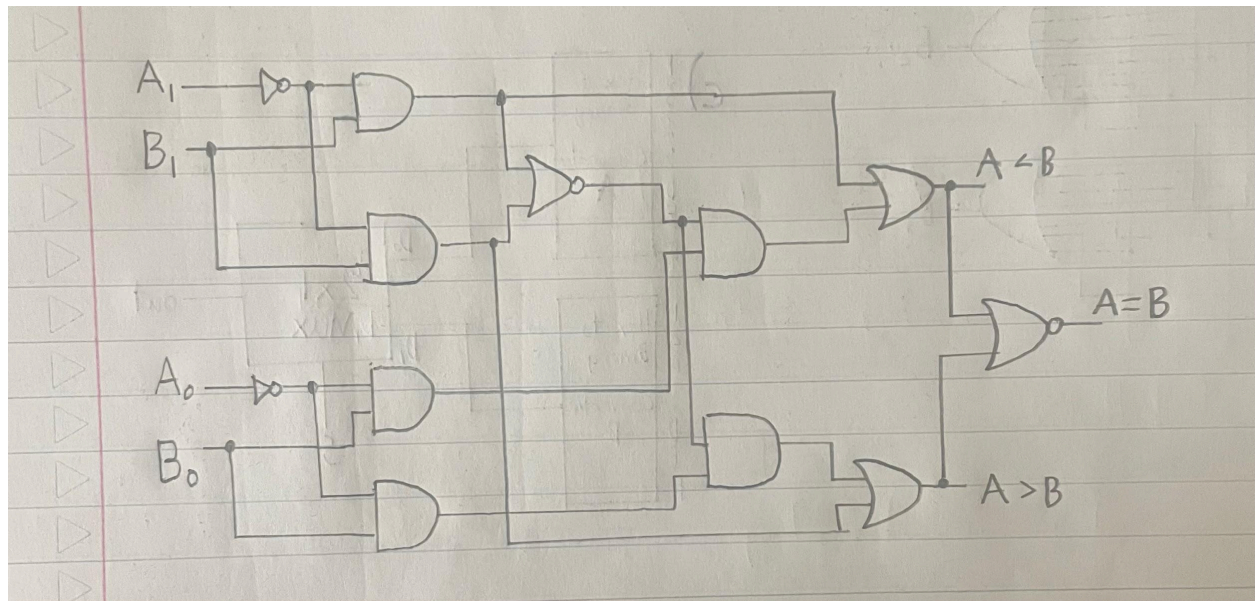
b)  $A = B$



c)  $A < B$

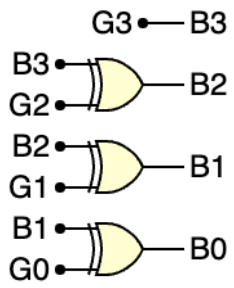


d)

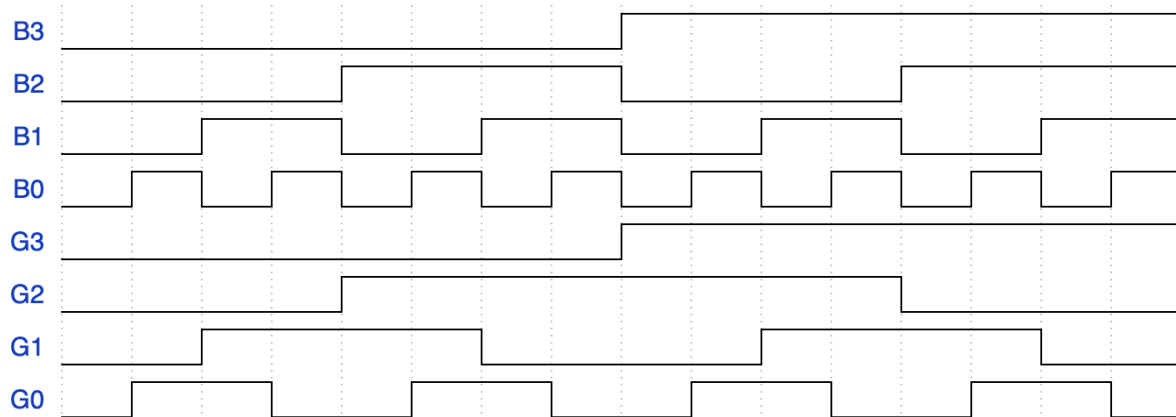


3.

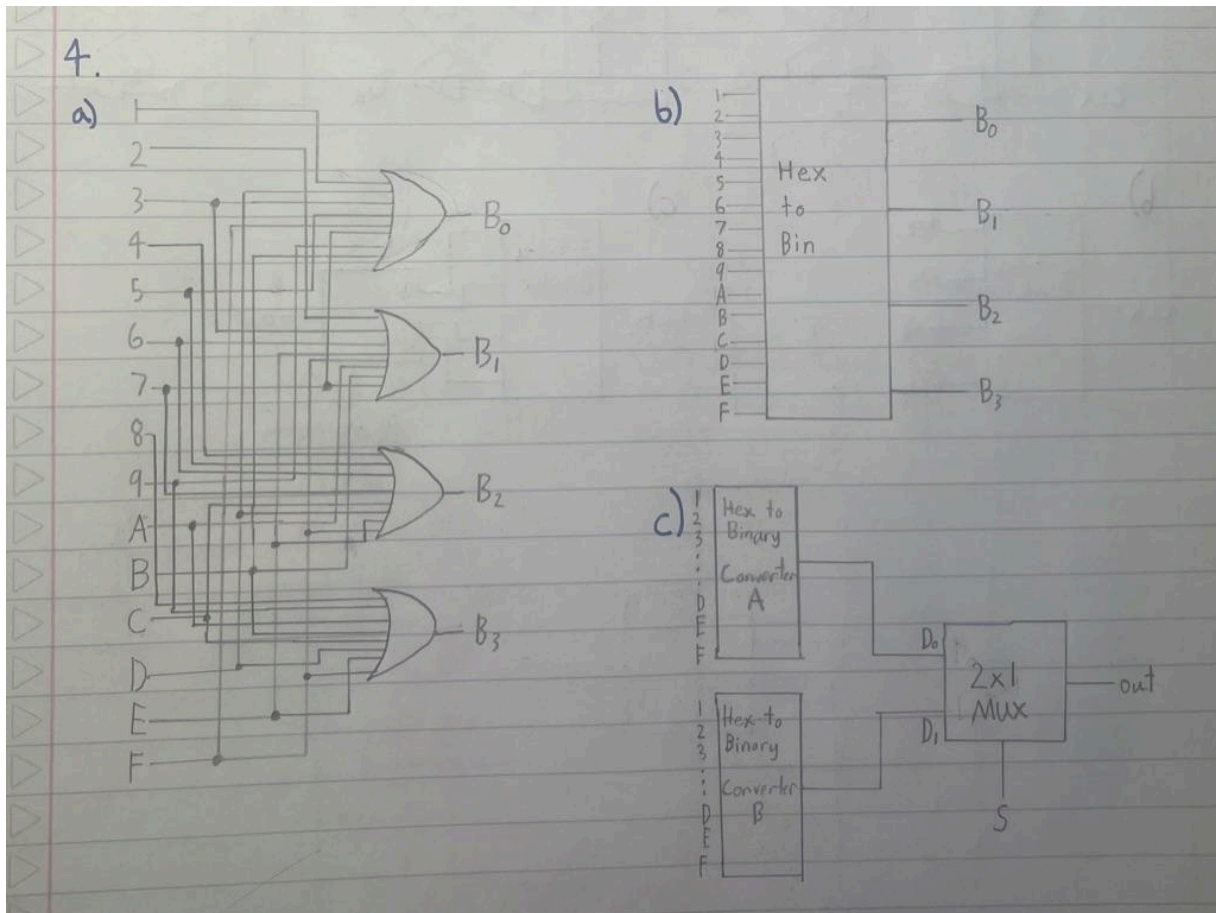
### Logic Gates



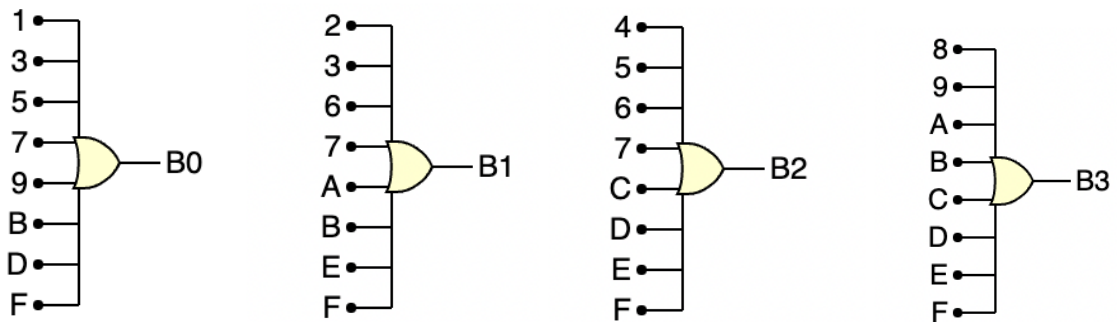
### Timing Diagram



4.



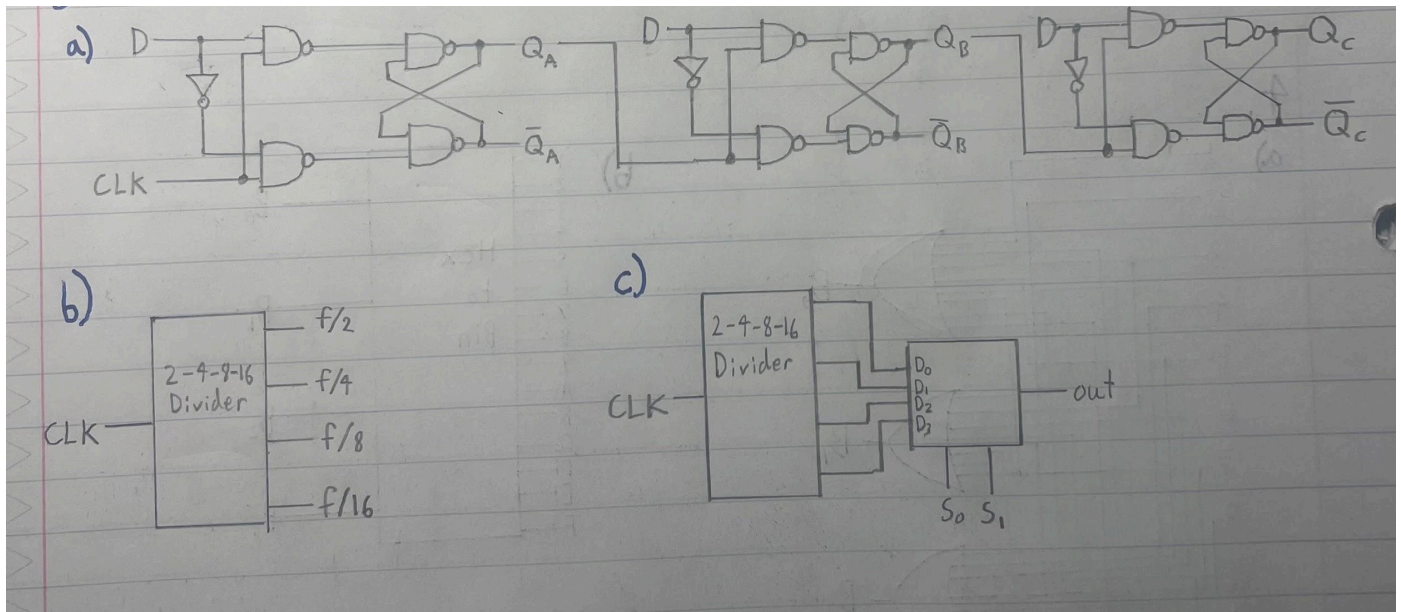
a)



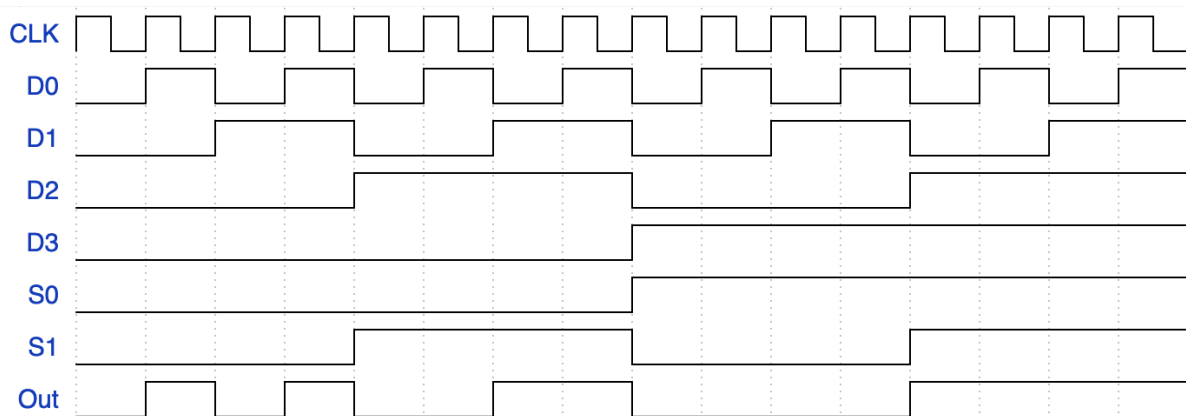


## Chapter 7

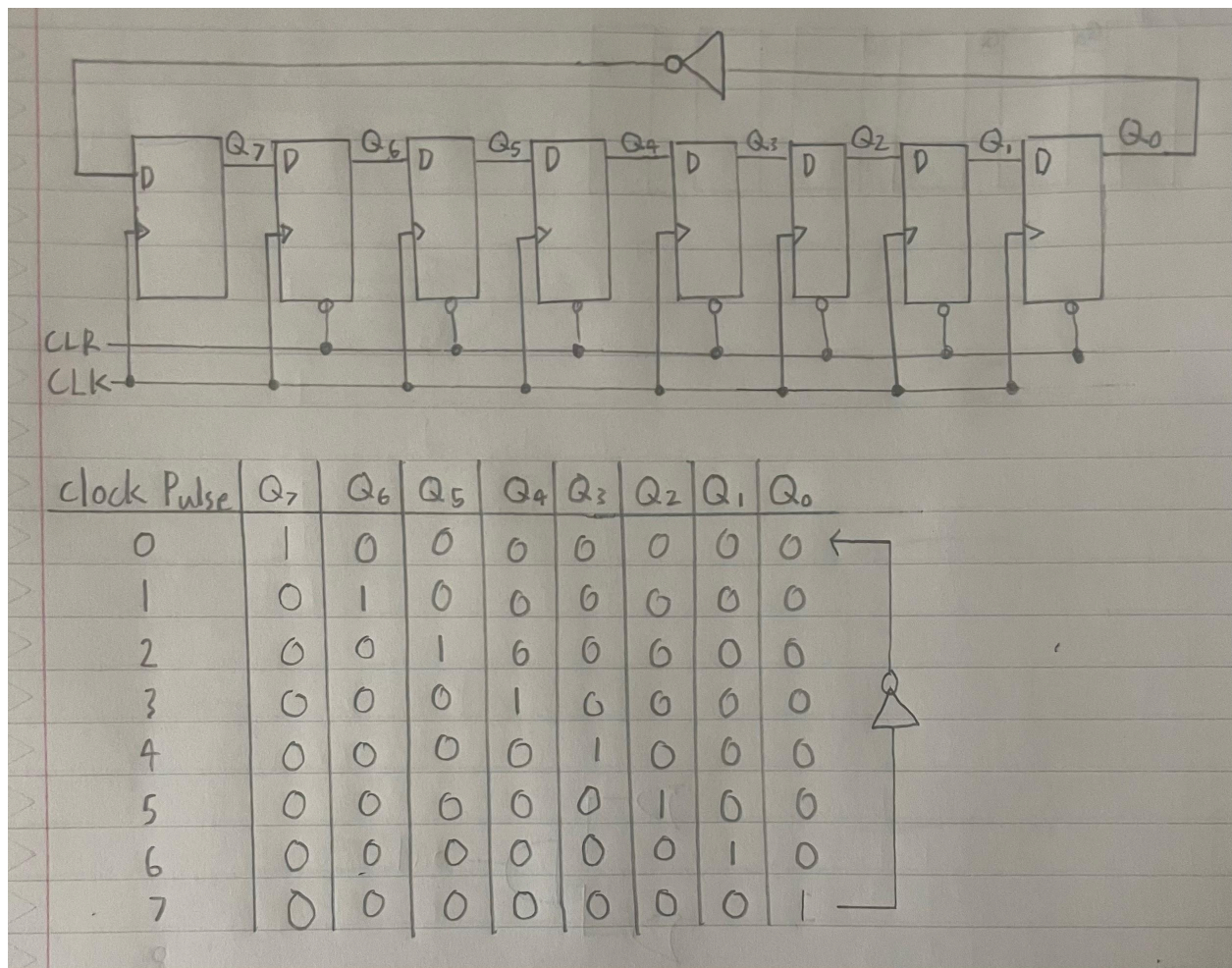
1.



d)

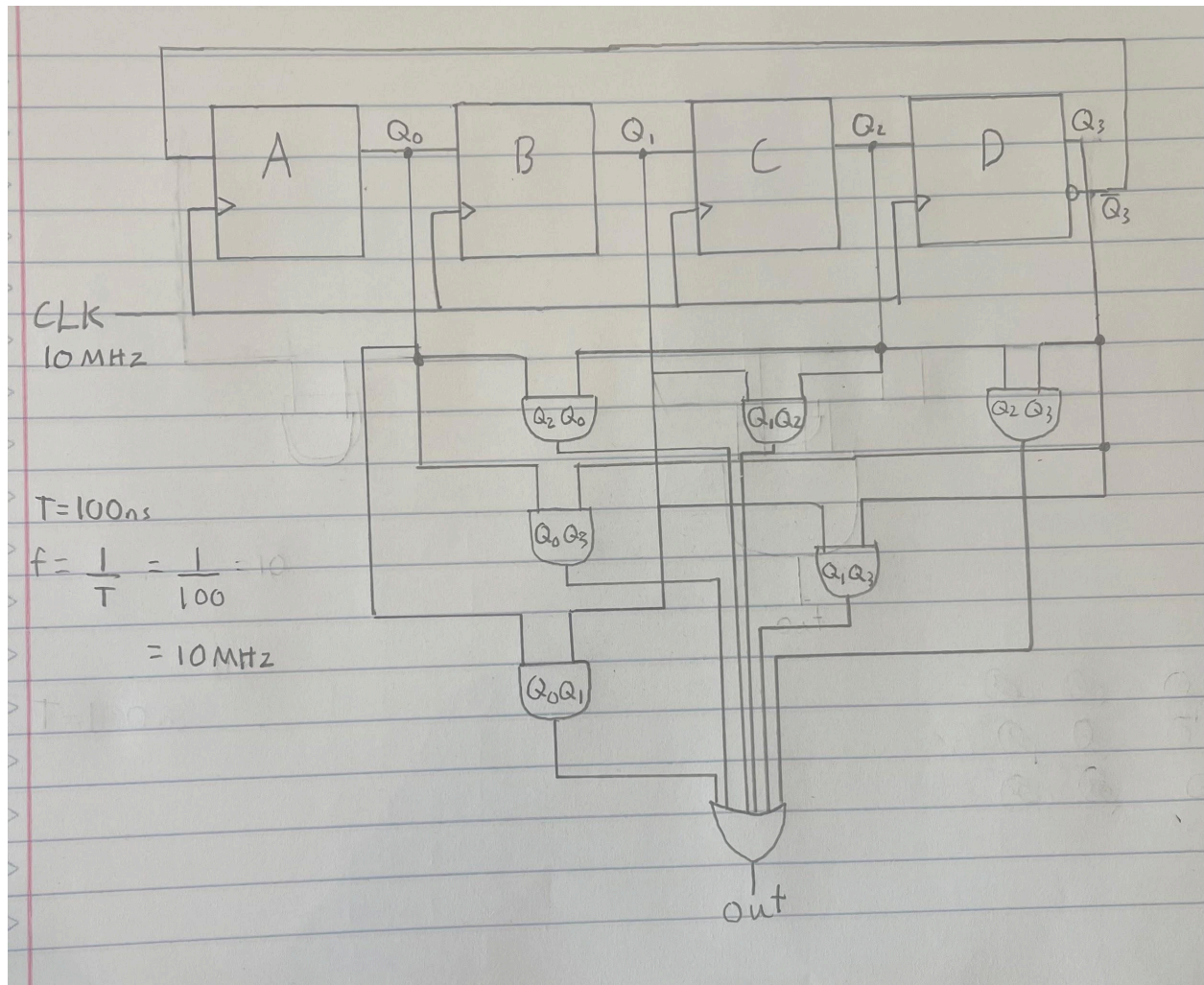


1.





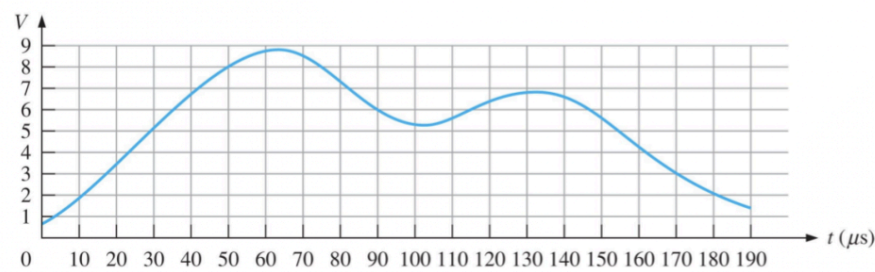
2.



Chapter 12

1.

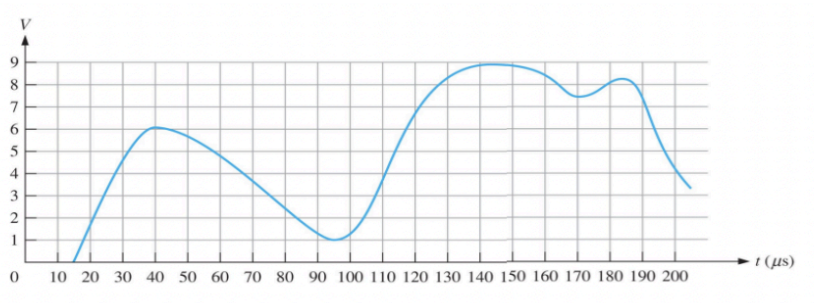
a)



|    |     |
|----|-----|
| 0  | 000 |
| 10 | 010 |
| 20 | 011 |
| 30 | 101 |
| 40 | 110 |
| 50 | 111 |
| 60 | 111 |
| 70 | 111 |
| 80 | 111 |
| 90 | 110 |

|     |     |
|-----|-----|
| 100 | 101 |
| 110 | 101 |
| 120 | 110 |
| 130 | 110 |
| 140 | 110 |
| 150 | 101 |
| 160 | 100 |
| 170 | 011 |
| 180 | 010 |
| 190 | 001 |

b)



|    |     |
|----|-----|
| 0  | 000 |
| 10 | 000 |
| 20 | 001 |
| 30 | 100 |
| 40 | 110 |
| 50 | 101 |
| 60 | 100 |
| 70 | 011 |
| 80 | 010 |
| 90 | 001 |

|     |     |
|-----|-----|
| 100 | 001 |
| 110 | 011 |
| 120 | 110 |
| 130 | 111 |
| 140 | 111 |
| 150 | 111 |
| 160 | 111 |
| 170 | 111 |
| 180 | 111 |
| 190 | 111 |

|     |     |
|-----|-----|
| 200 | 100 |
|-----|-----|



2.

2.

$$I_0 = \frac{5}{200} = 0.025 \text{ mA}$$

$$I_2 = \frac{5}{50} = 0.1 \text{ mA}$$

$$I_1 = \frac{5}{100} = 0.05 \text{ mA}$$

$$I_3 = \frac{5}{25} = 0.2 \text{ mA}$$

$$V_{out_0} = (10)(-0.025) = -0.25 \text{ V}$$

$$V_{out_2} = (10)(-0.1) = -1 \text{ V}$$

$$V_{out_1} = (10)(-0.05) = -0.5 \text{ V}$$

$$V_{out_3} = (10)(-0.2) = -2 \text{ V}$$

